

Exercices on session *Statistics with R - Multivariate analysis*

Load the packages *vegan* and *tidyverse*.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(vegan)
```

Loading required package: permute

Climate of the cities

In this exercise, you will study the climate of 49 big cities in the world. The data are in the *cities_climate.csv* file available [here](#).

- Load the data as a tibble and look at their structure. These data have been obtained from WorldClim. They contain:
 - long and lat: the longitude and latitude
 - t_mean: Mean annual temperature (°C)

- t_diu: Daily temperature variation, average difference between the maximum and minimum temperature in the same month (°C)
- t_sd: Seasonal temperature variation, standard deviation of the mean temperature between months (°C)
- t_max: Maximum temperature of the warmest month (°C)
- t_min: Minimum temperature of the coldest month (°C)
- p_ann: Annual precipitation (mm)
- p_max: Precipitation in the wettest month (mm)
- p_min: Precipitation of the driest month (mm)
- p_cv: Coefficient of variation of precipitation between months

```
dt_clim <- read_csv("data/cities_climate.csv")
```

```
Rows: 49 Columns: 12
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): city
```

```
dbl (11): long, lat, t_mean, t_diu, t_sd, t_max, t_min, p_ann, p_max, p_min,...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dt_clim
```

```
# A tibble: 49 x 12
```

	city	long	lat	t_mean	t_diu	t_sd	t_max	t_min	p_ann	p_max	p_min	p_cv
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Tokyo	140.	35.7	15.6	7.9	7.49	30.7	0.7	1428	185	47	40
2	Shanghai	121.	31.0	16.2	7.4	8.34	31.6	0.8	1050	167	41	46
3	Mumbai	72.8	19.0	26.8	7.3	1.77	32.7	18.9	2370	827	0	143
4	Karachi	67.1	24.9	25.9	9.9	4.04	34.2	12	194	82	0	134
5	Delhi	77.2	28.7	25.1	13.1	6.72	40.5	7.3	734	274	3	142
6	Manila	121.	14.6	27.3	8	1.08	33.8	21.4	2150	466	8	90
7	Moscow	37.6	55.8	4.9	7.7	9.80	23.3	-12.2	679	90	33	30
8	Seoul	127.	37.6	10.9	9.6	10.2	29	-9.5	1420	412	21	97
9	Sao Paulo	-46.7	-23.5	18.2	10	2.14	26.1	9.7	1388	233	34	59
10	Istanbul	29.0	41.0	14	7.4	6.29	27.4	3	759	126	24	49

```
# i 39 more rows
```

```
str(dt_clim)
```

```
spc_tbl_ [49 x 12] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ city   : chr [1:49] "Tokyo" "Shanghai" "Mumbai" "Karachi" ...
 $ long   : num [1:49] 139.8 121.4 72.8 67.1 77.2 ...
 $ lat    : num [1:49] 35.7 31 19 24.9 28.7 ...
 $ t_mean : num [1:49] 15.6 16.2 26.8 25.9 25.1 27.3 4.9 10.9 18.2 14 ...
 $ t_diu  : num [1:49] 7.9 7.4 7.3 9.9 13.1 8 7.7 9.6 10 7.4 ...
 $ t_sd   : num [1:49] 7.49 8.34 1.77 4.04 6.71 ...
 $ t_max  : num [1:49] 30.7 31.6 32.7 34.2 40.5 33.8 23.3 29 26.1 27.4 ...
 $ t_min  : num [1:49] 0.7 0.8 18.9 12 7.3 21.4 -12.2 -9.5 9.7 3 ...
 $ p_ann  : num [1:49] 1428 1050 2370 194 734 ...
 $ p_max  : num [1:49] 185 167 827 82 274 466 90 412 233 126 ...
 $ p_min  : num [1:49] 47 41 0 0 3 8 33 21 34 24 ...
 $ p_cv   : num [1:49] 40 46 143 134 142 90 30 97 59 49 ...
 - attr(*, "spec")=
 .. cols(
 ..   city = col_character(),
 ..   long = col_double(),
 ..   lat = col_double(),
 ..   t_mean = col_double(),
 ..   t_diu = col_double(),
 ..   t_sd = col_double(),
 ..   t_max = col_double(),
 ..   t_min = col_double(),
 ..   p_ann = col_double(),
 ..   p_max = col_double(),
 ..   p_min = col_double(),
 ..   p_cv = col_double()
 .. )
 - attr(*, "problems")=<pointer: 0x000001d5f3aff130>
```

We are going to do a PCA on these data.

- Remove the latitude and longitude

```
dt_clim <- dt_clim %>% select(-long, -lat)
```

- Put the variable *city* as a row names and remove the corresponding column (to be able to see the name of the cities in the output of the analysis).

```
dt_clim <- column_to_rownames(dt_clim, var = "city")

# as it has row names, it can no longer be a tibble

# alternatively, if the data were loaded as a dataframe and not a tibble
# dt_clim <- read.csv("data/cities_climate.csv")
# rownames(dt_clim) <- dt_clim$city
# dt_clim <- dt_clim[, -1]
```

- Do the PCA (don't forget to standardise the variables)

```
clim_pca <- pca(dt_clim, scale = TRUE)
```

- Look at the output: what is the cumulative proportion of variance explained by the two first axes?

```
summary(clim_pca)
```

Call:

```
pca(X = dt_clim, scale = TRUE)
```

Partitioning of correlations:

	Inertia	Proportion
Total	9	1
Unconstrained	9	1

Eigenvalues, and their contribution to the correlations

Importance of components:

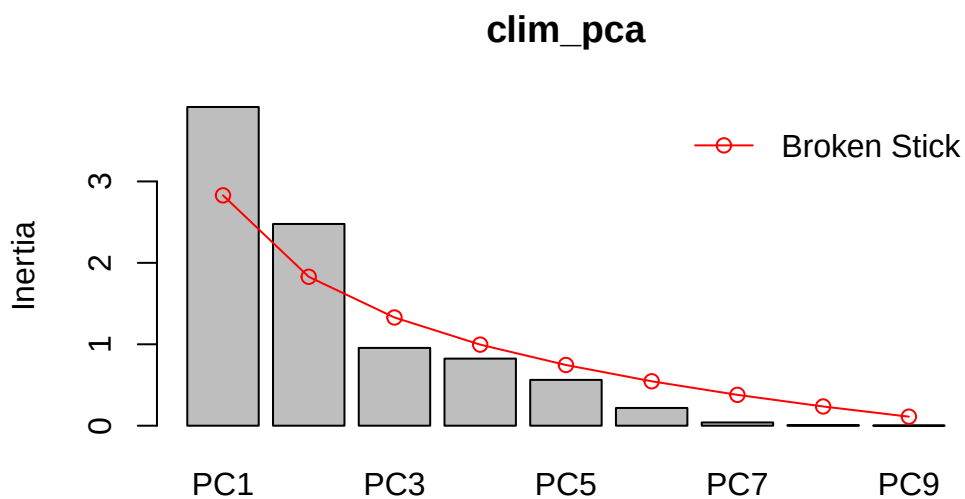
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Eigenvalue	3.9144	2.4778	0.9552	0.82386	0.56255	0.21766	0.040010
Proportion Explained	0.4349	0.2753	0.1061	0.09154	0.06251	0.02418	0.004446
Cumulative Proportion	0.4349	0.7102	0.8164	0.90792	0.97042	0.99461	0.999053

	PC8	PC9
Eigenvalue	0.0066269	0.0018966
Proportion Explained	0.0007363	0.0002107
Cumulative Proportion	0.9997893	1.0000000

```
# 71%
```

- Do a screeplot with a broken stick distribution: how many axes do you want to interpret?

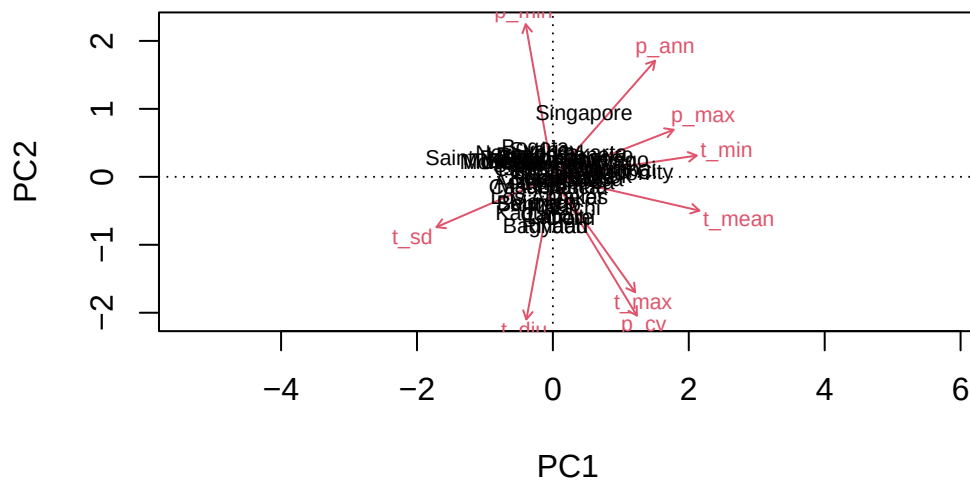
```
screeplot(clim_pca, bstick = TRUE)
```



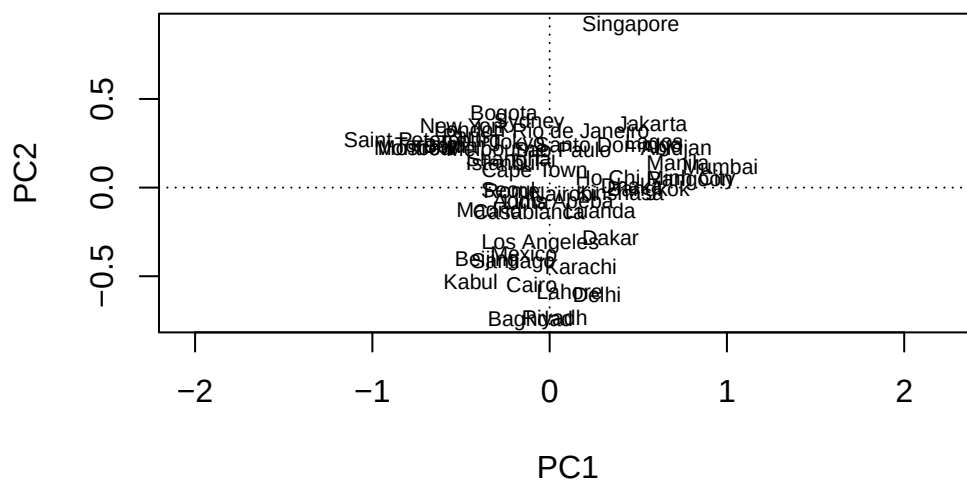
```
# 2 first axes
```

- Let's first look at similarity between cities: which scaling do we need for the biplot? Do the biplot and interpret it.

```
biplot(clim_pca, scaling = 1) # scaling 1 focus on the observations
```

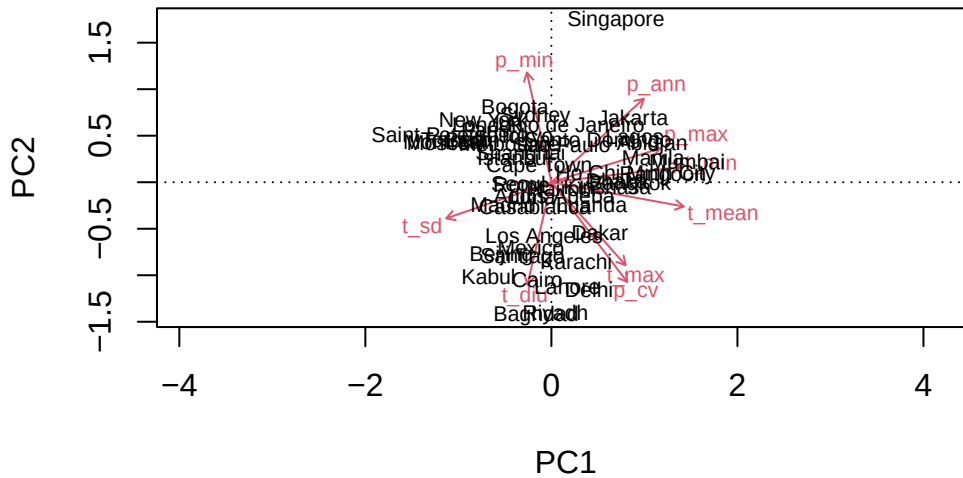


```
# to see better, show only the cities
biplot(clim_pca, scaling = 1, display = "sites")
```

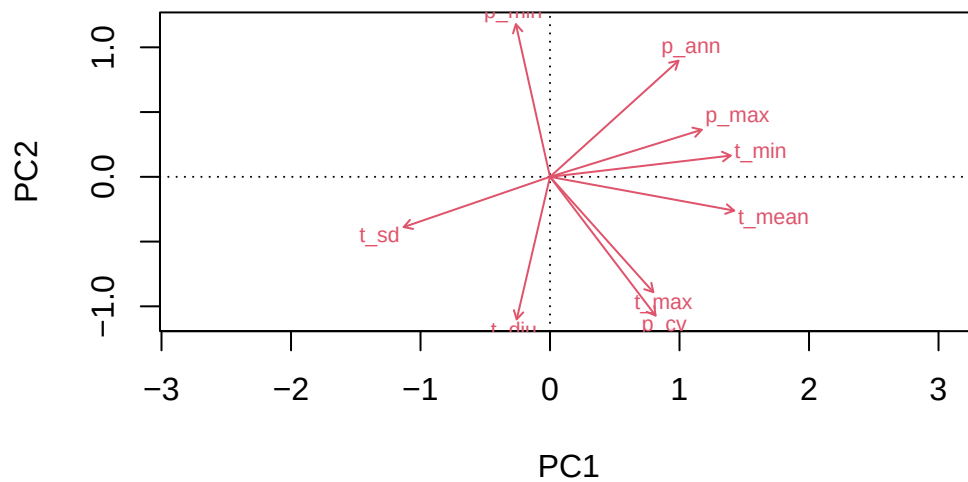


- Let's now look at correlations between climatic variables: do the corresponding biplot and interpret the axes.

```
biplot(clim_pca, scaling = 2) # scaling 2 focuses on the variables
```



```
# to see better, show only the variables
biplot(clim_pca, scaling = 2, display = "species")
```



```
# interpretation
# first axis: from warm (right) to cold (left)
# 2nd axis: from warm with high seasonality (bottom ) to cold, less seasonal
```

- Let's now prepare the data for a clustering: standardise them and compute the distance matrix.

```
dt_clim_sd <- as.data.frame(scale(dt_clim))
clim_dist <- dist(dt_clim_sd)
```

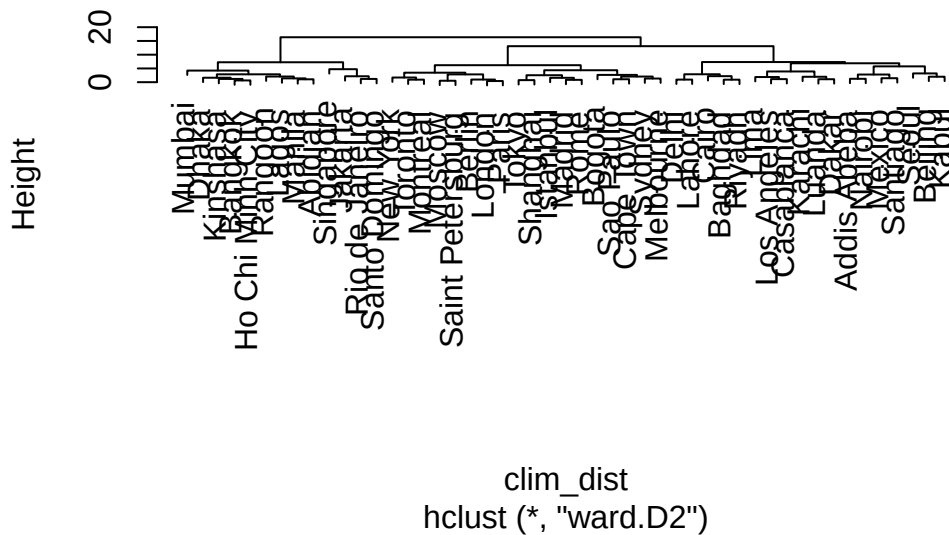
- Do the hierarchical clustering using the method of Ward.

```
clim_clust <- hclust(clim_dist, method = "ward.D2")
```

- Plot the dendrogram.

```
plot(clim_clust)
```

Cluster Dendrogram



- Make 3 groups and extract them.

```
city_3gr <- cutree(clim_clust, k = 3)
```

- Represent the groups on the PCA biplot.

```
dt_city_gr <- as.data.frame(scores(clim_pca,
                                   scaling = "sites",
                                   display = "sites",
                                   choices = 1:2))

# check that plots are in the same order
all(rownames(city_3gr) == rownames(city_3gr))
```

```
[1] TRUE
```

```
# add groups to the PCs data
dt_city_gr$groups <- as.factor(city_3gr)

# plot
ggplot(dt_city_gr, aes(x = PC1, y = PC2, colour = groups)) +
  geom_text(aes(label = rownames(dt_city_gr)), size = 3) +
  theme_bw()
```

